22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA
S2 B.Tech CSE-AI
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus
CASE STUDY
PHASE – I (Analysis)
Problem Identification & Class Mapping (CO1)

Group No : 5
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1. Problem Title :

Online Examination System

2. Problem Description :

Traditional examination systems involve manual question paper preparation, physical attendance, and time-consuming evaluation processes. This leads to delays, human errors, and inefficiency.

The Online Examination System aims to digitize the entire process by enabling students to take exams online, allowing faculty to create and manage questions, and enabling automatic evaluation by the system. This ensures faster result generation, improved accuracy, and reduced manual workload.

3. Identified Entities (Classes):

Class Name	Description
User	Base class for all system users
Student	Represents a student taking exams
Faculty	Represents a faculty member who creates questions
Admin	Manages scheduling and system operations
Exam	Represents an examination
Question	Represents a question in an exam
Result	Stores exam results

EvaluationSystem	Handles automatic evaluation
Authentication	Manages login and security
Answer	Stores student responses

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
User (Abstract)	userId, name, email,	login(), logout()
Student	studentId,	takeExam(), viewResult()
Faculty	facultyId, subjects	Represents a faculty member who
Admin	adminId	scheduleExam(), manageUsers()
Exam	examID, title, duration, questionsList, scheduledTime	startExam(), endExam()
Question(Abstract)	questionId, questionText, marks	displayQuestion(), checkAnswer() (abstract)
MCQQuestion	options[], correctOption	checkAnswer()
DescriptiveQuestion	sampleAnswer	checkAnswer()
Answer	answerID, studentResponse, questionId	submitAnswer()
Result	resultId, studentId, score, grade	calculateGrade(), displayResult()
EvaluationSystem	evaluationId	evaluateExam(), calculateScore()
Authentication	authId	validateUser(), generateSession()